Greedy Algorithms

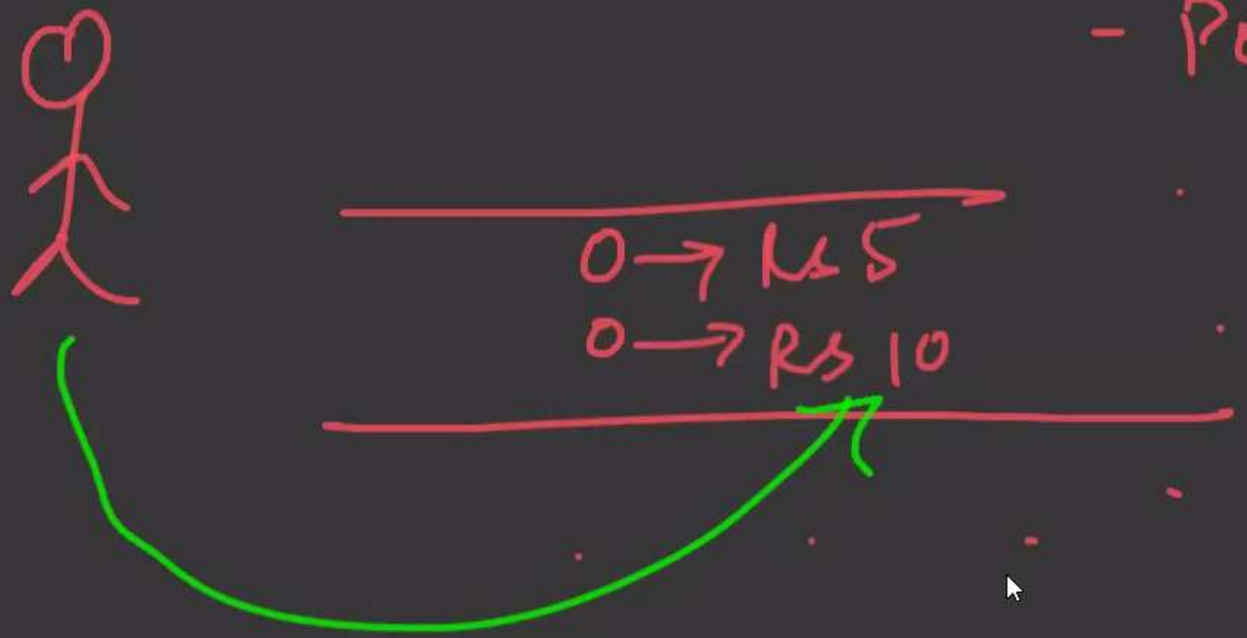
Dev Karan Singh (devkaran1231)
Expert at codeforces (1817)
5 star at codechef (2041)

Greedy Algorithms

- What are Greedy Algorithms?
- Kadane's Algorithm
- Lexicographical order of Strings (1 question)
- Questions :
  - Find Original Array From Doubled Array
  - https://codeforces.com/problemset/problem/33/A
  - https://codeforces.com/problemset/problem/620/C
  - https://codeforces.com/problemset/problem/587/A
  - https://codeforces.com/problemset/problem/723/C

greedy Algos:

Eg:

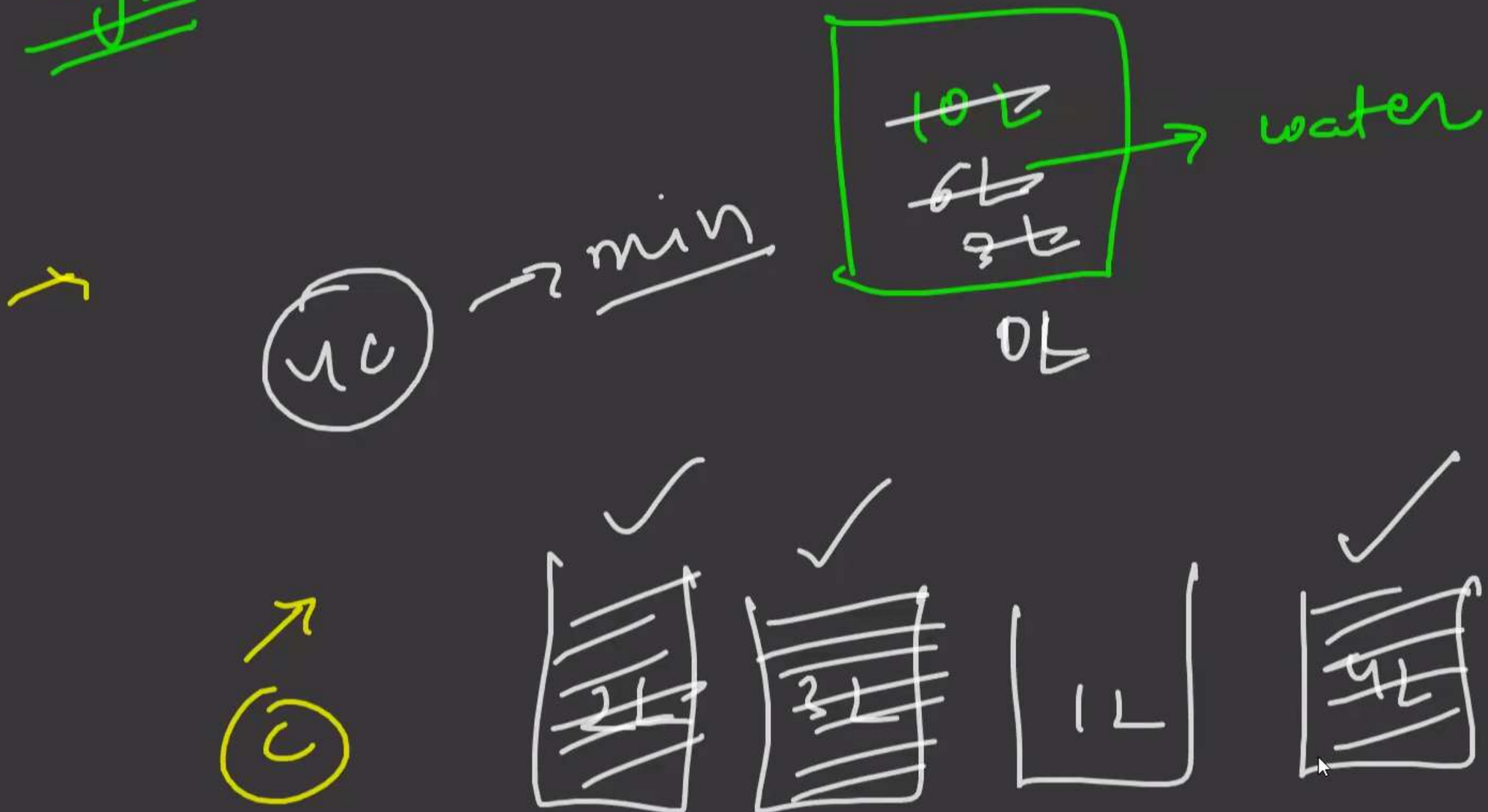


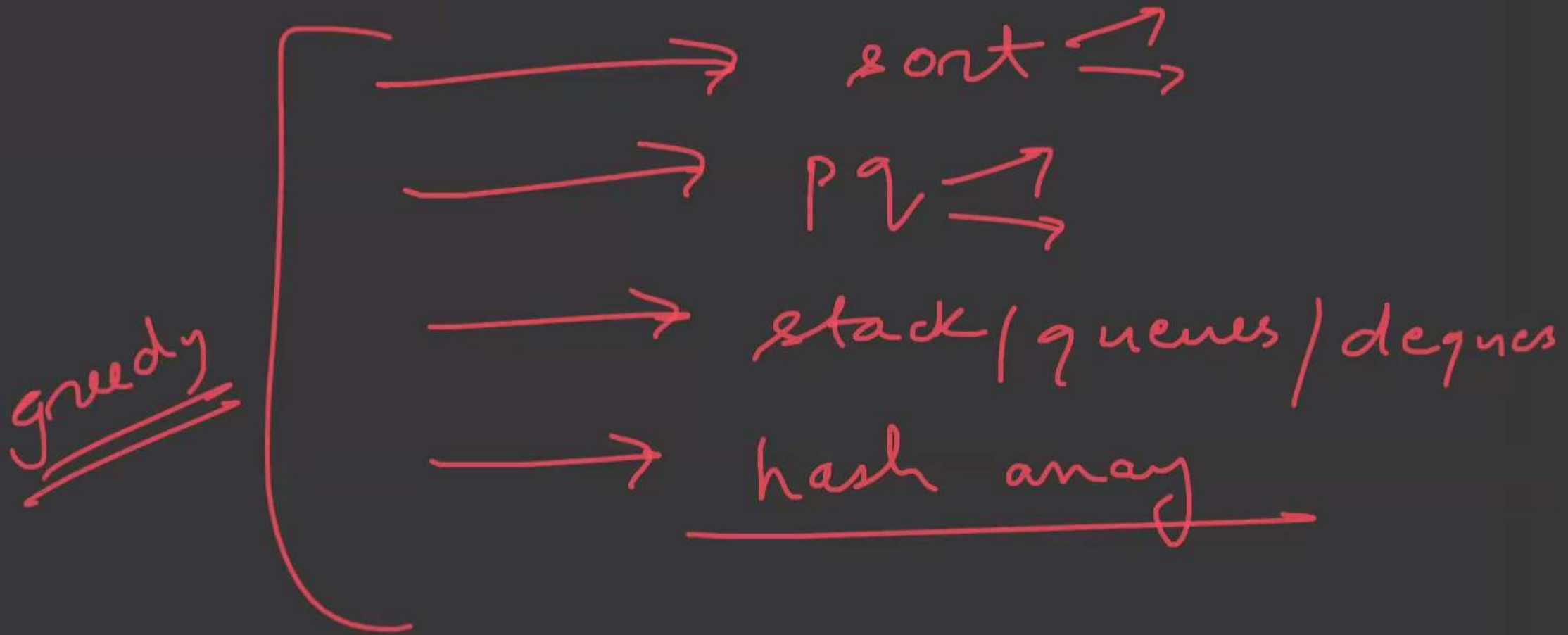
Once

1 sec

odeforces.com/problemset/problem/620/C

Eg:




greedy
sort
pq
stack / queues / deques
hash array

Kadane's Algo:

Q → given an array, find the subarray with max sum.

n = 7

Eg:

4 2 -7 3 5 -3 5

odeforces.com/problemset/problem/620/C

4 2 -7

3 5 - 3 5

-1

3+0

sum (new) → (3) → 3

sum (count) → (3 - 1) → 2

Eg: -1 3 5 -3 5

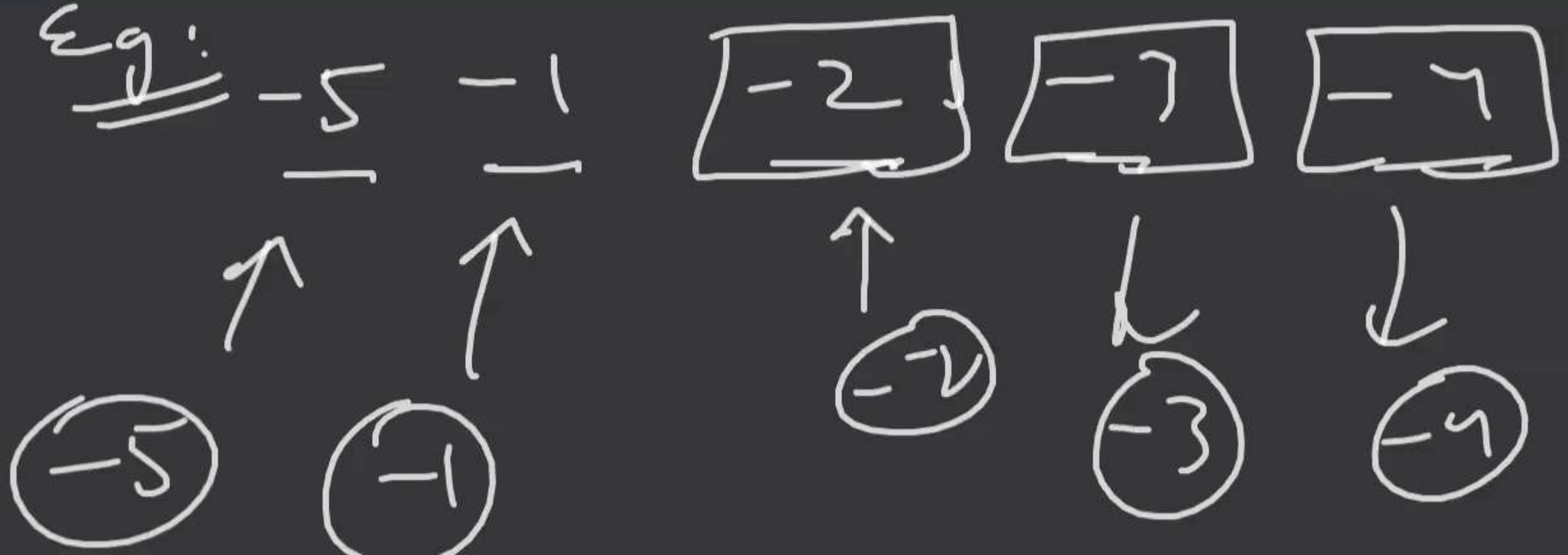
sum → -ve → resultant

↓

-ve

sum ≥ 0

Loss ⟵ leave it


Eg:
-5
-1
-2
-3
-4
-5
-1
-2
-3
-4

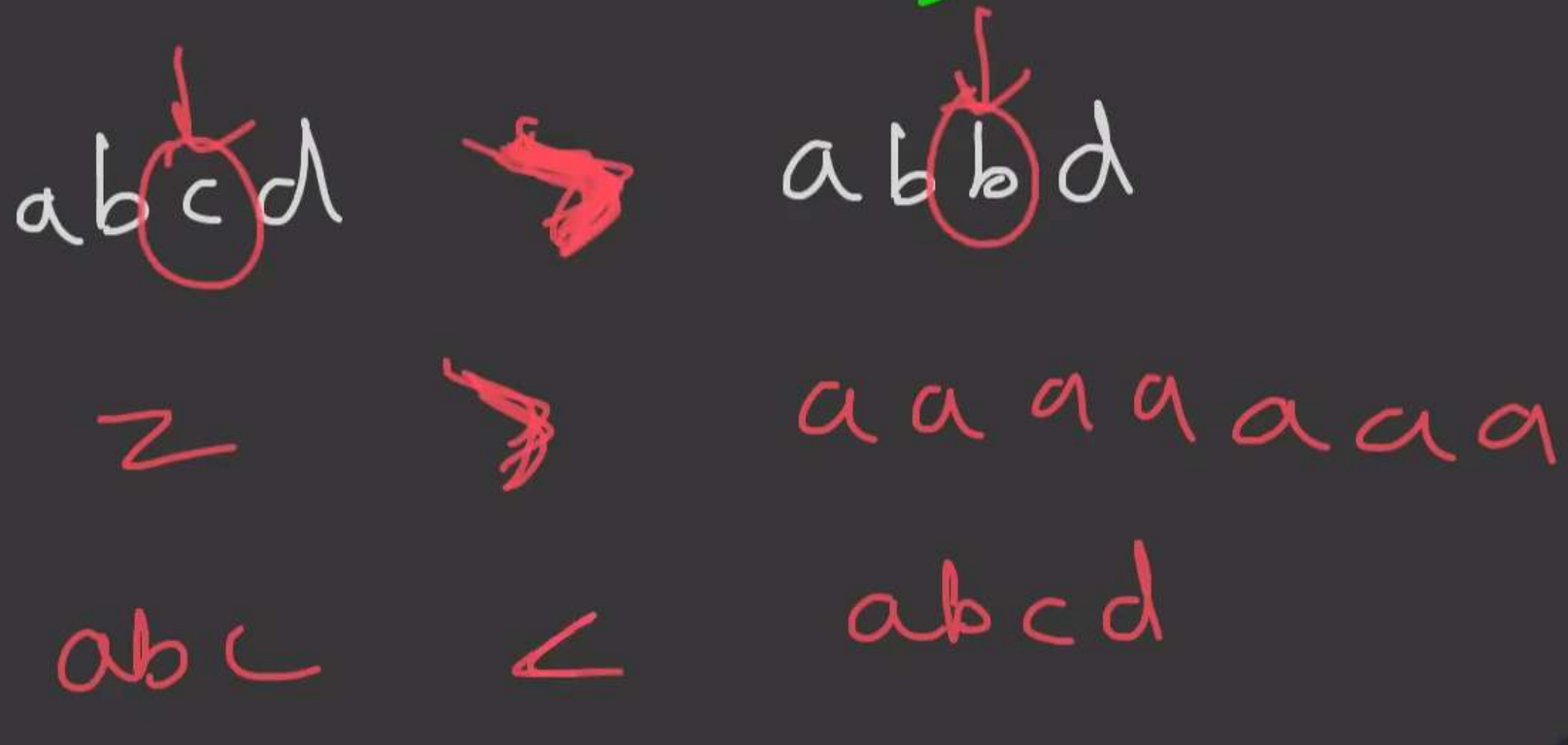
Eg: -5 -1 -2 -7 -6

sum = 0

max → ~~-∞~~ -5 -1

Lexicographical order :) (In strings)

→ a b c d . . . — x y z



Q string given (all lowercase).
Rearrange the string to get
lex. min string

condition → adj elements should
differ more than
k.

eg:

azz

k = 2

→ zaz ✓

→ azz ✗

→ zza

devkaran

a → 2
d → 1
e → 1
k → 1
n → 1
r → 1
v → 1

a d a e k n n v

a → 2 + 0

d → 1 0

e → 1 0

l ← → 1 0

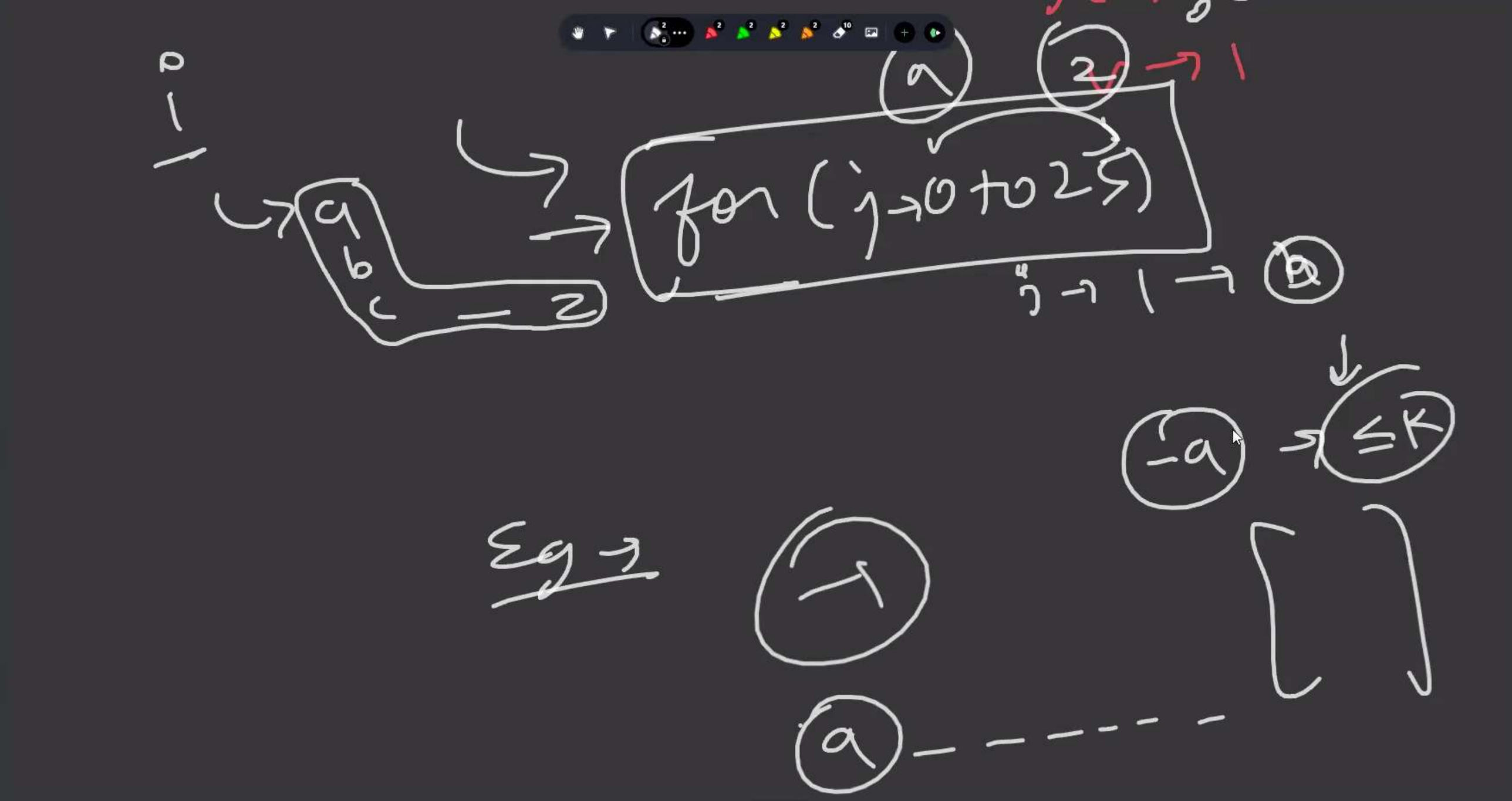
n → 1 ←

y → ~~y~~ ←

v → 1

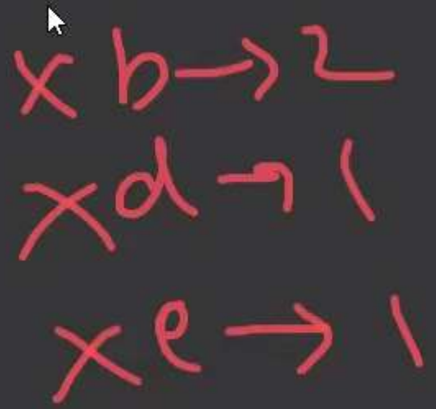
for (i → 0 to n)

for (j → 0 to 26)


a
2
for (j→0 to 25)
j → 1 → a
Eg →
-a
≤K




a


x b → 2
x d → 1
x e → 1


K = S

a d

x b → 2

x e → 1

K = 2

-1

$\rightarrow$ ~~X~~ c ~~g~~ h

$K = 2$

$a \in h \subseteq \mathcal{F}$

→ a c e g h

K = 2

a c e g h

h

h

a c e g

backtracking